

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01  
AND EXPERT SYSTEMS

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### COURSE OUTCOME COVERED

**CO3:** *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)

Assessment Tool Weightage: 15%

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**Total Marks: 100**

**Time: 60 Mins**

**No. of Questions: 5**

Annexure A

### Logic Reasoning & Mapping Exercise

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#### **Code of Conduct:**

*I G. Poojitha (192425337) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: G. Poojitha**

| Criterion                                           | Weightage | Marks Obtained |
|-----------------------------------------------------|-----------|----------------|
| 1. Problem Analysis and Knowledge Identification    | 15%       |                |
| 2. Logical Representation and Mapping               | 20%       |                |
| 3. Predicates, Variables, Constants and Quantifiers | 15%       |                |
| 4. Logical Connectives and Formula Construction     | 10%       |                |
| 5. Inference and Logical Reasoning                  | 15%       |                |
| 6. Unification and Substitution                     | 10%       |                |
| 7. Resolution and Proof Construction                | 10%       |                |
| 8. Mapping Diagram / Clarity and Presentation       | 5%        |                |
| Total                                               | 100%      |                |

### ANSWER ANY ONE OF THE FOLLOWING

| Q.No | Use Case Scenario                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Smart Healthcare Diagnosis</b> | <p>A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention.</p> <p>Tasks:</p> <ul style="list-style-type: none"> <li>(i) Identify facts, entities, predicates, variables, and constants.</li> <li>(ii) Represent the knowledge using Propositional Logic.</li> <li>(iii) Represent the knowledge using FOL with suitable quantifiers.</li> <li>(iv) Perform unification for two relevant predicates.</li> <li>(v) Apply resolution to prove whether Ravi has a viral infection.</li> <li>(vi) Show the complete inference sequence and final conclusion.</li> </ul>                                                    |
| 2    | <b>Student Academic Advising</b>  | <p>A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support.</p> <p>Tasks:</p> <ul style="list-style-type: none"> <li>(i) Identify the knowledge components.</li> <li>(ii) Convert the statements into Propositional Logic.</li> <li>(iii) Develop FOL representations using predicates and quantifiers.</li> <li>(iv) Identify suitable substitutions and perform unification.</li> <li>(v) Use resolution to prove whether Priya requires academic support.</li> <li>(vi) Explain the reasoning steps leading to the conclusion.</li> </ul> |
| 3    | <b>Intelligent Loan Approval</b>  | <p>A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval.</p> <p>Tasks:</p> <ul style="list-style-type: none"> <li>(i) Identify entities, facts, predicates, and rules.</li> <li>(ii) Map the scenario into Propositional Logic.</li> <li>(iii) Represent the rules using FOL.</li> </ul>                                                                                                                                                             |

|   |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                       | <p>(iv) Demonstrate unification and substitution.</p> <p>(v) Use resolution to prove or disprove LoanApproved(Arun).</p> <p>(vi) Provide the final inference with justification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4 | <b>Vehicle Fault Diagnosis</b>        | <p>An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service.</p> <p>Tasks:</p> <p>(i) Extract the facts and rules from the scenario.</p> <p>(ii) Represent them using Propositional Logic.</p> <p>(iii) Develop suitable FOL predicates and rules.</p> <p>(iv) Perform unification for the vehicle-related predicates.</p> <p>(v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.</p>                                                                              |
| 5 | <b>Cybersecurity Threat Detection</b> | <p>An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach.</p> <p>Tasks:</p> <p>(i) Identify the facts, entities, predicates, and rules.</p> <p>(ii) Map the knowledge into Propositional Logic.</p> <p>(iii) Represent the rules using FOL.</p> <p>(iv) Demonstrate unification and substitution.</p> <p>(v) Apply resolution to determine whether User101 is involved in a possible security breach.</p> <p>(vi) Present the complete reasoning chain and final inference.</p> |

## EVALUATION RUBRICS:

| Criterion                                        | Weightage (%) | Excellent                                                                                                                    | Good                                                                 | Average                                                                  | Poor                                                            |
|--------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------|
| Problem Understanding & Knowledge Identification | 15%           | 13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario. | 10–12% – Identifies most relevant knowledge with minor omissions.    | 7–9% – Identifies basic knowledge but misses several important elements. | 0–6% – Poor or incorrect identification of the given knowledge. |
| Logical Representation & Mapping                 | 20%           | 17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.         | 13–16% – Provides mostly correct logical mappings with minor errors. | 9–12% – Provides partial mappings with several logical errors.           | 0–8% – Logical mapping is largely incorrect or missing.         |
| Predicates, Variables, Constants & Quantifiers   | 15%           | 13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.           | 10–12% – Uses the concepts correctly with minor errors.              | 7–9% – Demonstrates partial understanding with some incorrect usage.     | 0–6% – Concepts are incorrectly applied or missing.             |
| Logical Connectives & Formula Construction       | 10%           | 9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to                                                      | 7–8% – Mostly correct use of connectives with minor errors.          | 5–6% – Basic use of connectives but several errors occur.                | 0–4% – Incorrect or incomplete formula construction.            |

|                                         |     |                                                                                                                |                                                                     |                                                                    |                                                               |
|-----------------------------------------|-----|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|
|                                         |     | construct valid logical formulas.                                                                              |                                                                     |                                                                    |                                                               |
| Inference & Logical Reasoning           | 15% | 13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.      | 10–12% – Reasoning is mostly correct with minor gaps.               | 7–9% – Provides partial reasoning with some incorrect conclusions. | 0–6% – Reasoning is incorrect, unsupported, or missing.       |
| Unification & Substitution              | 10% | 9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.         | 7–8% – Correctly performs most unification steps with minor errors. | 5–6% – Demonstrates basic understanding but has several errors.    | 0–4% – Unable to perform unification correctly.               |
| Resolution / Proof Construction         | 10% | 9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.                 | 7–8% – Resolution is mostly correct with minor errors.              | 5–6% – Shows partial understanding but proof contains gaps.        | 0–4% – Resolution is incorrect or not demonstrated.           |
| Clarity, Mapping Diagram & Presentation | 5%  | 5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation. | 4% – Presentation is clear with minor formatting or diagram issues. | 3% – Basic presentation with limited clarity.                      | 0–2% – Poorly organized, unclear, or incomplete presentation. |

# LOAN INTELLIGENT APPROVAL SYSTEM

## CHAPTER 1: PROBLEM ANALYSIS & KNOWLEDGE IDENTIFICATION

### 1.1 Scenario Overview & Automated Credit Risk Context

Modern commercial banking institutions rely heavily on AI-driven Knowledge-Based Systems (KBS) to execute preliminary loan processing and automated risk classification. Automated reasoning engines process applicant attribute data against codified underwriting guidelines to deliver consistent, transparent, and legally auditable credit decisions. In this assessment scenario, an automated loan approval system is tasked with evaluating the creditworthiness of a applicant named Arun.

The operational process follows a multi-tiered conditional inference model. Raw financial facts (employment stability, income adequacy, credit bureau score, and historical repayment performance) are incrementally synthesized into intermediate risk evaluations (preliminary approval qualification and low-risk risk-tier assignment) before reaching a final recommendation on loan approval.

### 1.2 System Entities, Domain Constants, and Variables

Knowledge representation requires distinguishing between concrete domain objects (constants) and placeholder domain abstractions (variables):

- 1. Primary Constant Entity — 'Arun':** Represents the specific individual applicant currently undergoing loan risk evaluation. In formal logic syntax, 'Arun' is designated as a 0-arity constant symbol.
- 2. Universally Quantified Domain Variable — 'x':** Represents any arbitrary customer or loan applicant belonging to the universal domain of bank clients. Universal rules quantified over x apply to all applicants in the database.

**3. Risk Tier & Status Entities:** Abstract organizational classes within the bank's decision framework, specifically 'Preliminary Approval Status', 'Low-Risk Tier Classification', and 'Loan Approval Recommendation'.

1.3 Fact Extraction & Empirical Fact Assertions

| Fact ID | Natural Language Assertion         | Predicate Logic Notation      | Verification Source & Meaning                   |
|---------|------------------------------------|-------------------------------|-------------------------------------------------|
| Fact F1 | Arun has stable employment.        | HasStableEmployment(Arun)     | Employer verification confirmed > 2 yrs tenure. |
| Fact F2 | Arun has sufficient income.        | HasSufficientIncome(Arun)     | Debt-to-income ratio < 35% verified.            |
| Fact F3 | Arun has a high credit score.      | HasHighCreditScore(Arun)      | Credit bureau score > 750 points.               |
| Fact F4 | Arun has a good repayment history. | HasGoodRepaymentHistory(Arun) | Zero defaults across prior 36 months.           |

1.4 Domain Rules & Logical Axioms

| Rule ID | Underwriting Guidelines Statement                                                        | Inference Level        | Formal Premise-Consequent Logic                                            |
|---------|------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------|
| Rule R1 | Customers with stable employment and sufficient income qualify for preliminary approval. | Stage 1: Qualification | $(\text{Stable} \wedge \text{Income}) \rightarrow \text{PrelimApproved}$   |
| Rule R2 | Customers with a high credit score and preliminary approval are classified as low-risk.  | Stage 2: Risk Tiering  | $(\text{Credit} \wedge \text{PrelimApproved}) \rightarrow \text{LowRisk}$  |
| Rule R3 | Low-risk customers with good repayment history can be recommended for loan approval.     | Stage 3: Final Action  | $(\text{LowRisk} \wedge \text{Repayment}) \rightarrow \text{LoanApproved}$ |

# CHAPTER 2: PROPOSITIONAL LOGIC REPRESENTATION & MAPPING

## 2.1 Propositional Symbol Definitions

In Propositional Logic, atomic sentences are abstracted as simple propositional symbols that hold boolean truth values (True or False). We define seven formal proposition symbols for Arun's loan evaluation:

| Symbol | Proposition Description                  | Assigned Value in Scenario |
|--------|------------------------------------------|----------------------------|
| P      | Arun has stable employment.              | TRUE (Fact F1)             |
| Q      | Arun has sufficient income.              | TRUE (Fact F2)             |
| R      | Arun has a high credit score.            | TRUE (Fact F3)             |
| S      | Arun has a good repayment history.       | TRUE (Fact F4)             |
| A      | Arun qualifies for preliminary approval. | Derived Intermediate       |
| L      | Arun is classified as low-risk.          | Derived Intermediate       |
| W      | Arun is recommended for loan approval.   | Target Query Goal          |

## 2.2 Propositional Knowledge Base Structure

### PROPOSITIONAL KNOWLEDGE BASE (KB<sub>prop</sub>)

Atomic Fact Assertions:

- Fact F1: P (Arun has stable employment)
- Fact F2: Q (Arun has sufficient income)



- Fact F3: R (Arun has a high credit score)
- Fact F4: S (Arun has a good repayment history)

Implication Rules:

- Rule R1\_prop:  $(P \wedge Q) \rightarrow A$  (Stable employment AND sufficient income IMPLIES preliminary approval)
- Rule R2\_prop:  $(R \wedge A) \rightarrow L$  (High credit score AND preliminary approval IMPLIES low-risk tier)
- Rule R3\_prop:  $(L \wedge S) \rightarrow W$  (Low-risk tier AND good repayment history IMPLIES loan approval)

### 2.3 Propositional Truth Table & Model-Checking Analysis

To prove that the Knowledge Base entails the final approval recommendation ( $KB\_prop \models W$ ), we conduct model checking across truth assignments. A model is an assignment of True/False to all variables that satisfies every statement in  $KB\_prop$ . Out of  $2^7 = 128$  possible truth assignments, exactly 1 model satisfies all background facts and rules:

| P | Q | R | S | $A = (P \wedge Q)$ | $L = (R \wedge A)$ | $W = (L \wedge S)$ | KB Satisfiability |
|---|---|---|---|--------------------|--------------------|--------------------|-------------------|
| T | T | T | T | T                  | T                  | T                  | VALID MODEL       |
| T | T | T | T | T                  | T                  | F                  | CONTRADICTION     |

## CHAPTER 3: FIRST-ORDER LOGIC (FOL) FORMALIZATION

### 3.1 Syntax, Predicates, and Universal Quantifiers

First-Order Logic (FOL) extends propositional logic by introducing terms, predicates, and quantifiers. Unary predicates define properties of domain entities. Below are the seven primary predicates used in our formulation:

| Predicate Symbol           | Arity & Type | Semantic Interpretation in Banking Domain                             |
|----------------------------|--------------|-----------------------------------------------------------------------|
| HasStableEmployment(x)     | Unary (1)    | Evaluates to True if client x has stable employment.                  |
| HasSufficientIncome(x)     | Unary (1)    | Evaluates to True if client x's income meets requirements.            |
| HasHighCreditScore(x)      | Unary (1)    | Evaluates to True if client x's credit score is high.                 |
| HasGoodRepaymentHistory(x) | Unary (1)    | Evaluates to True if client x has no history of default.              |
| PreliminaryApproval(x)     | Unary (1)    | Evaluates to True if client x qualifies for preliminary approval.     |
| LowRisk(x)                 | Unary (1)    | Evaluates to True if client x is classified as low-risk.              |
| LoanApproved(x)            | Unary (1)    | Evaluates to True if client x is recommended for final loan approval. |

### 3.2 Ground Facts and Quantified Rules in First-Order Logic

The complete First-Order Knowledge Base KB\_FOL is composed of four ground atomic facts and three universally quantified rules:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>FIRST-ORDER LOGIC KNOWLEDGE BASE (KB_FOL)</b></p> <p>Ground Fact Statements (Ground Atomic Formulas):</p> <ul style="list-style-type: none"> <li>• Fact F1: HasStableEmployment(Arun)</li> <li>• Fact F2: HasSufficientIncome(Arun)</li> <li>• Fact F3: HasHighCreditScore(Arun)</li> <li>• Fact F4: HasGoodRepaymentHistory(Arun)</li> </ul> <p>Universally Quantified Domain Rules:</p> <ul style="list-style-type: none"> <li>• Rule R1: <math>\forall x [ ( \text{HasStableEmployment}(x) \wedge \text{HasSufficientIncome}(x) ) \rightarrow \text{PreliminaryApproval}(x) ]</math></li> <li>• Rule R2: <math>\forall x [ ( \text{HasHighCreditScore}(x) \wedge \text{PreliminaryApproval}(x) ) \rightarrow \text{LowRisk}(x) ]</math></li> <li>• Rule R3: <math>\forall x [ ( \text{LowRisk}(x) \wedge \text{HasGoodRepaymentHistory}(x) ) \rightarrow \text{LoanApproved}(x) ]</math></li> </ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## CHAPTER 4: INFERENCE & LOGICAL REASONING STEPS

### 4.1 Forward Chaining Derivation Trace

Forward chaining is a data-driven inference process that starts with facts in the Knowledge Base and applies Modus Ponens repeatedly to derive new assertions until the target conclusion is derived.

#### Stage 1: Derivation of Preliminary Approval Status

1. Premises Available: Fact F1 = HasStableEmployment(Arun), Fact F2 = HasSufficientIncome(Arun).
2. Universal Instantiation: Instantiate Rule R1 with substitution  $\{x / \text{Arun}\}$ :  
 $(\text{HasStableEmployment}(\text{Arun}) \wedge \text{HasSufficientIncome}(\text{Arun})) \rightarrow \text{PreliminaryApproval}(\text{Arun})$ .
3. Conjunction Introduction: Combine F1 and F2 to form  $(\text{HasStableEmployment}(\text{Arun}) \wedge \text{HasSufficientIncome}(\text{Arun}))$ .
4. Modus Ponens Application: From the conjunction and instantiated Rule R1, derive:  
Derived Fact DF1: PreliminaryApproval(Arun).

#### Stage 2: Derivation of Low-Risk Classification

1. Premises Available: Fact F3 = HasHighCreditScore(Arun), Derived Fact DF1 = PreliminaryApproval(Arun).
2. Universal Instantiation: Instantiate Rule R2 with substitution  $\{x / \text{Arun}\}$ :  
 $(\text{HasHighCreditScore}(\text{Arun}) \wedge \text{PreliminaryApproval}(\text{Arun})) \rightarrow \text{LowRisk}(\text{Arun})$ .
3. Conjunction Introduction: Combine F3 and DF1 to form  $(\text{HasHighCreditScore}(\text{Arun}) \wedge \text{PreliminaryApproval}(\text{Arun}))$ .
4. Modus Ponens Application: From the conjunction and instantiated Rule R2, derive:  
Derived Fact DF2: LowRisk(Arun).

#### Stage 3: Derivation of Final Loan Approval

1. Premises Available: Fact F4 = HasGoodRepaymentHistory(Arun), Derived Fact DF2 = LowRisk(Arun).
2. Universal Instantiation: Instantiate Rule R3 with substitution  $\{x / \text{Arun}\}$ :  
 $(\text{LowRisk}(\text{Arun}) \wedge \text{HasGoodRepaymentHistory}(\text{Arun})) \rightarrow \text{LoanApproved}(\text{Arun})$ .

3. Conjunction Introduction: Combine DF2 and F4 to form  $(\text{LowRisk}(\text{Arun}) \wedge \text{HasGoodRepaymentHistory}(\text{Arun}))$ .

4. Modus Ponens Application: Derive Final Conclusion:

Final Conclusion:  $\text{LoanApproved}(\text{Arun})$ .

## 4.2 Backward Chaining Goal Reduction Trace

Backward chaining works backward from the query goal ? -  $\text{LoanApproved}(\text{Arun})$  to establish supporting facts:

- **Step 1 (Initial Goal):** Goal =  $\text{LoanApproved}(\text{Arun})$ . Matches Rule R3 consequent with substitution  $\{x / \text{Arun}\}$ . New Subgoals:  $\text{LowRisk}(\text{Arun})$  AND  $\text{HasGoodRepaymentHistory}(\text{Arun})$ .
- **Step 2 (Fact Verification & Subgoal Reduction):** Subgoal  $\text{HasGoodRepaymentHistory}(\text{Arun})$  directly matches Fact F4. Remaining Subgoal:  $\text{LowRisk}(\text{Arun})$ .
- **Step 3 (Rule R2 Matching):** Subgoal  $\text{LowRisk}(\text{Arun})$  matches Rule R2 consequent with  $\{x / \text{Arun}\}$ . New Subgoals:  $\text{HasHighCreditScore}(\text{Arun})$  AND  $\text{PreliminaryApproval}(\text{Arun})$ .
- **Step 4 (Fact Verification & Rule R1 Matching):** Subgoal  $\text{HasHighCreditScore}(\text{Arun})$  directly matches Fact F3. Subgoal  $\text{PreliminaryApproval}(\text{Arun})$  matches Rule R1 consequent with  $\{x / \text{Arun}\}$ , generating Subgoals:  $\text{HasStableEmployment}(\text{Arun})$  AND  $\text{HasSufficientIncome}(\text{Arun})$ .
- **Step 5 (Final Grounding):** Both  $\text{HasStableEmployment}(\text{Arun})$  and  $\text{HasSufficientIncome}(\text{Arun})$  match Facts F1 and F2. All leaf nodes are grounded in KB facts. Proof successful!

# CHAPTER 5: UNIFICATION ALGORITHM & SUBSTITUTION MECHANISM

## 5.1 Unification Principles and MGU Theory

Unification is the process of finding a substitution  $\theta = \{v_1/t_1, \dots, v_n/t_n\}$  that makes two logical expressions syntactically identical. The Most General Unifier (MGU) minimizes variable constraints. Unification requires checking predicate symbols, arity equality, and executing the Occurs Check to prevent circular variable assignments.

## 5.2 Detailed Algorithmic Step Traces for Predicate Unification

### Unification Problem 1: Unifying Universal Rule Premise with Ground Fact F1

Expression 1:  $P1 = \text{HasStableEmployment}(x)$

Expression 2:  $P2 = \text{HasStableEmployment}(\text{Arun})$

Trace Steps:

1. Predicate Symbol Check: 'HasStableEmployment' == 'HasStableEmployment'. Symbols match.
2. Arity Check:  $\text{Arity}(P1) = 1, \text{Arity}(P2) = 1$ . Arities match.
3. Argument Term Comparison: Term 1 = Variable  $x$ , Term 2 = Constant Arun.
4. Occurs Check: Variable  $x$  does not occur in constant Arun. Check passes.
5. Substitution Derivation:  $\theta_1 = \{ x / \text{Arun} \}$ .
6. Verification:  $P1 \theta_1 = \text{HasStableEmployment}(\text{Arun}) = P2$ . Unification successful.

### Unification Problem 2: Unifying Rule R2 Consequent with LowRisk Query

Expression 1:  $P1 = \text{LowRisk}(y)$

Expression 2:  $P2 = \text{LowRisk}(\text{Arun})$

Trace Steps:

1. Predicate Symbol Check: 'LowRisk' == 'LowRisk'. Match successful.
2. Arity Check:  $1 = 1$ . Match successful.

3. Argument Binding: Bind variable y to constant Arun.
4. Substitution Derivation:  $\theta_2 = \{ y / \text{Arun} \}$ .
5. Verification:  $P1 \theta_2 = \text{LowRisk}(\text{Arun}) == P2$ . Unification successful.

### Unification Algorithm Execution Summary

| Pair ID | Expression 1 (Pattern) | Expression 2 (Target)     | MGU ( $\theta$ )      | Unified Ground Result     |
|---------|------------------------|---------------------------|-----------------------|---------------------------|
| U1      | HasStableEmployment(x) | HasStableEmployment(Arun) | $\{x / \text{Arun}\}$ | HasStableEmployment(Arun) |
| U2      | PreliminaryApproval(x) | PreliminaryApproval(Arun) | $\{x / \text{Arun}\}$ | PreliminaryApproval(Arun) |
| U3      | LowRisk(y)             | LowRisk(Arun)             | $\{y / \text{Arun}\}$ | LowRisk(Arun)             |

## CHAPTER 6: RESOLUTION REFUTATION PROOF CONSTRUCTION

### 6.1 Conversion of Knowledge Base to Conjunctive Normal Form (CNF)

To perform Resolution Refutation, all logical formulas in the Knowledge Base must be converted into Conjunctive Normal Form (CNF), where each formula is a conjunction of disjunctive clauses. The transformation involves: (1) Eliminating material implication ( $A \rightarrow B \equiv \neg A \vee B$ ), (2) Standardizing variables across clauses, and (3) Dropping universal quantifiers.

#### Step-by-Step Transformation of Rule Axioms to CNF Clauses:

- Rule R1 Transformation:

$$\begin{aligned}
 & \forall x [ (\text{HasStableEmployment}(x) \wedge \text{HasSufficientIncome}(x)) \rightarrow \text{PreliminaryApproval}(x) ] \\
 & \equiv \forall x [ \neg(\text{HasStableEmployment}(x) \wedge \text{HasSufficientIncome}(x)) \vee \text{PreliminaryApproval}(x) ] \\
 & \equiv \forall x [ \neg\text{HasStableEmployment}(x) \vee \neg\text{HasSufficientIncome}(x) \vee \text{PreliminaryApproval}(x) ]
 \end{aligned}$$

Clause C1:  $\{ \neg \text{HasStableEmployment}(x1) \vee \neg \text{HasSufficientIncome}(x1) \vee \text{PreliminaryApproval}(x1) \}$

• Rule R2 Transformation:

$\forall x [ (\text{HasHighCreditScore}(x) \wedge \text{PreliminaryApproval}(x)) \rightarrow \text{LowRisk}(x) ]$

$\equiv \forall x [ \neg \text{HasHighCreditScore}(x) \vee \neg \text{PreliminaryApproval}(x) \vee \text{LowRisk}(x) ]$

Clause C2:  $\{ \neg \text{HasHighCreditScore}(x2) \vee \neg \text{PreliminaryApproval}(x2) \vee \text{LowRisk}(x2) \}$

• Rule R3 Transformation:

$\forall x [ (\text{LowRisk}(x) \wedge \text{HasGoodRepaymentHistory}(x)) \rightarrow \text{LoanApproved}(x) ]$

$\equiv \forall x [ \neg \text{LowRisk}(x) \vee \neg \text{HasGoodRepaymentHistory}(x) \vee \text{LoanApproved}(x) ]$

Clause C3:  $\{ \neg \text{LowRisk}(x3) \vee \neg \text{HasGoodRepaymentHistory}(x3) \vee \text{LoanApproved}(x3) \}$

• Facts F1-F4 CNF Clauses:

Clause C4:  $\{ \text{HasStableEmployment}(\text{Arun}) \}$

Clause C5:  $\{ \text{HasSufficientIncome}(\text{Arun}) \}$

Clause C6:  $\{ \text{HasHighCreditScore}(\text{Arun}) \}$

Clause C7:  $\{ \text{HasGoodRepaymentHistory}(\text{Arun}) \}$

• Negated Goal Target Clause:

Target Query:  $\text{LoanApproved}(\text{Arun})$

Negated Query Assertion Clause C8:  $\{ \neg \text{LoanApproved}(\text{Arun}) \}$

## 6.2 Master Clause Set Table

| Clause ID | Origin Source      | CNF Disjunctive Clause Representation                                                                              |
|-----------|--------------------|--------------------------------------------------------------------------------------------------------------------|
| C1        | Rule R1 (Prelim)   | $\neg \text{HasStableEmployment}(x1) \vee \neg \text{HasSufficientIncome}(x1) \vee \text{PreliminaryApproval}(x1)$ |
| C2        | Rule R2 (Low-Risk) | $\neg \text{HasHighCreditScore}(x2) \vee \neg \text{PreliminaryApproval}(x2) \vee \text{LowRisk}(x2)$              |
| C3        | Rule R3 (Approval) | $\neg \text{LowRisk}(x3) \vee \neg \text{HasGoodRepaymentHistory}(x3) \vee \text{LoanApproved}(x3)$                |

|    |                        |                               |
|----|------------------------|-------------------------------|
| C4 | Fact F1 (Employment)   | HasStableEmployment(Arun)     |
| C5 | Fact F2 (Income)       | HasSufficientIncome(Arun)     |
| C6 | Fact F3 (Credit Score) | HasHighCreditScore(Arun)      |
| C7 | Fact F4 (Repayment)    | HasGoodRepaymentHistory(Arun) |
| C8 | Negated Goal           | $\neg$ LoanApproved(Arun)     |

### 6.3 Step-by-Step Resolution Proof Table

Resolution refutation repeatedly resolves pairs of clauses with complementary literals until the empty clause ( $\square$  or FALSE) is produced:

| Step   | Parent 1 | Parent 2 | Substitution $\theta$  | Literals Resolved                                                     | Derived Resolvent Clause                                                      |
|--------|----------|----------|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Step 1 | C3       | C8       | $\{x3 / \text{Arun}\}$ | LoanApproved(Arun) vs $\neg$ LoanApproved(Arun)                       | C9: $\neg$ LowRisk(Arun) $\vee$ $\neg$ HasGoodRepaymentHistory(Arun)          |
| Step 2 | C9       | C7       | None                   | $\neg$ HasGoodRepaymentHistory(Arun) vs HasGoodRepaymentHistory(Arun) | C10: $\neg$ LowRisk(Arun)                                                     |
| Step 3 | C2       | C10      | $\{x2 / \text{Arun}\}$ | LowRisk(Arun) vs $\neg$ LowRisk(Arun)                                 | C11: $\neg$ HasHighCreditScore(Arun) $\vee$ $\neg$ PreliminaryApproval(Arun)  |
| Step 4 | C11      | C6       | None                   | $\neg$ HasHighCreditScore(Arun) vs HasHighCreditScore(Arun)           | C12: $\neg$ PreliminaryApproval(Arun)                                         |
| Step 5 | C1       | C12      | $\{x1 / \text{Arun}\}$ | PreliminaryApproval(Arun) vs $\neg$ PreliminaryApproval(Arun)         | C13: $\neg$ HasStableEmployment(Arun) $\vee$ $\neg$ HasSufficientIncome(Arun) |
| Step 6 | C13      | C4       | None                   | $\neg$ HasStableEmployment(Arun) vs HasStableEmployment(Arun)         | C14: $\neg$ HasSufficientIncome(Arun)                                         |



|        |     |    |      |                                                                                             |                                          |
|--------|-----|----|------|---------------------------------------------------------------------------------------------|------------------------------------------|
| Step 7 | C14 | C5 | None | $\neg \text{HasSufficientIncome}(\text{Arun})$ vs $\text{HasSufficientIncome}(\text{Arun})$ | $\square$ (EMPTY CLAUSE / CONTRADICTION) |
|--------|-----|----|------|---------------------------------------------------------------------------------------------|------------------------------------------|

**PROOF CONCLUSION & FORMAL VERIFICATION**

- Derivation of Empty Clause ( $\square$ ): Step 7 produces a formal contradiction between C14 and C5.
- Mathematical Significance: The assumption that  $\neg \text{LoanApproved}(\text{Arun})$  is FALSE.
- Final Assertion:  $\text{LoanApproved}(\text{Arun})$  IS PROVED TO BE TRUE WITH 100% MATHEMATICAL CERTAINTY.

**CHAPTER 7: INTEGRATED LOGIC FLOWCHART & EVALUATION SUMMARY**

**7.1 Decision Logic Flowchart Architecture**

Below is the complete end-to-end decision workflow presented in a clear, self-contained visual callout structure that strictly avoids splitting across page boundaries:

**STRUCTURED LOGIC FLOWCHART: LOAN INTELLIGENT APPROVAL SYSTEM**

[ START: Customer Loan Application Initiated for Arun ]



[ INPUT DATA VERIFICATION & GROUND FACTS ]

├─ Fact F1:  $\text{HasStableEmployment}(\text{Arun}) == \text{TRUE}$

├─ Fact F2:  $\text{HasSufficientIncome}(\text{Arun}) == \text{TRUE}$

├─ Fact F3:  $\text{HasHighCreditScore}(\text{Arun}) == \text{TRUE}$

└─ Fact F4:  $\text{HasGoodRepaymentHistory}(\text{Arun}) == \text{TRUE}$



[ EVALUATION STAGE 1: PRELIMINARY APPROVAL ]

├ Rule R1: (Stable Employment  $\wedge$  Sufficient Income)  $\rightarrow$  Preliminary Approval

├ Unification: {x / Arun}

└ Result: PreliminaryApproval(Arun) = TRUE [Derived Fact DF1]

|



[ EVALUATION STAGE 2: RISK TIER CLASSIFICATION ]

├ Rule R2: (High Credit Score  $\wedge$  Preliminary Approval)  $\rightarrow$  Low-Risk Class

├ Unification: {x / Arun}

└ Result: LowRisk(Arun) = TRUE [Derived Fact DF2]

|



[ EVALUATION STAGE 3: FINAL RECOMMENDATION ]

├ Rule R3: (Low-Risk Class  $\wedge$  Good Repayment History)  $\rightarrow$  Loan Approved

├ Unification: {x / Arun}

└ Result: LoanApproved(Arun) = TRUE [Final Conclusion]

|



[ RESOLUTION REFUTATION VERIFICATION ]

├ Negated Query:  $\neg$ LoanApproved(Arun) [Clause C8]

├ Iterative Literal Elimination Steps 1 through 7

└ Result: Derivation of Empty Clause  $\square$  (Contradiction Established)

|



[ END: LOAN APPROVED FOR ARUN - NOTIFICATION ISSUED ]